MATHEMATICS ASSIGNMENT: PERCENTAGES

Instructor: Saksham Virmani **Topic:** Percentages

Name: _____ **Date:** _____ **Score:** _____

Instructions: Read each question carefully. Show all necessary calculations and write your final answer on the line provided.

Question 1 PERCENTAGE OF A QUANTITY

A water storage tank has a maximum capacity of 650 liters. Currently, it is filled to 72% of its total capacity. How many liters of water are currently in the tank?

Answer: _____

Question 2 FINDING PERCENTAGE SHARE

In a shipment of 800 ceramic tiles, inspection reveals that 56 tiles are cracked. What percentage of the tiles in the shipment are cracked?

Answer: _____

Question 3 PERCENTAGE INCREASE / EXPANSION

A metallic rod measures 140 cm at room temperature. When heated, its length expands by 3.5%. What is the new length of the rod after heating?

Answer: _____

Question 4 MULTI-PART COMPOSITION

In a sample of soil weighing 450 grams, 18% is moisture, 42% is sand, and the remainder consists of organic matter. What is the total mass of organic matter in the sample?

Answer: _____

Question 5 LENGTH PARTITIONING

A cable measuring 180 meters was cut into two pieces. The first piece represents 35% of the total original length.

- a) What is the length of the first piece?
- b) What is the length of the second piece?

a) _____ **b)** _____